

Day 7 Activity: ggplot2 internals with storms

Wednesday, May 27, 2026

Your Name Here

Setup

```
library(tidyverse)
```

We'll use the `storms` dataset that comes built into the tidyverse (via `dplyr`). It contains 6-hourly observations of every named Atlantic storm since 1975, from NOAA's HURDAT2 reanalysis.

```
glimpse(storms)
```

```
Rows: 20,778
Columns: 13
 $ name      <chr> "Amy", "Amy", "Amy", "Amy", "Amy", "Amy", ~
 $ year      <dbl> 1975, 1975, 1975, 1975, 1975, 1975, 1975, ~
 $ month     <dbl> 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, ~
 $ day       <int> 27, 27, 27, 27, 28, 28, 28, 28, 29, 29, 2~
 $ hour      <dbl> 0, 6, 12, 18, 0, 6, 12, 18, 0, 6, 12, 18, ~
 $ lat       <dbl> 27.5, 28.5, 29.5, 30.5, 31.5, 32.4, 33.3, ~
 $ long      <dbl> -79.0, -79.0, -79.0, -79.0, -78.8, -~
 78.7, ~
 $ status    <fct> tropical depression, tropical depression, ~
 $ category  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
 $ wind      <int> 25, 25, 25, 25, 25, 25, 25, 30, 35, 40, 4~
 $ pressure  <int> 1013, 1013, 1013, 1013, 1012, 1012, 1011, ~
 $ tropicalstorm_force_diameter <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
 $ hurricane_force_diameter    <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
```

Useful variables for today:

- name, year, month, day, hour — when the observation was taken
- lat, long — where the storm was
- status — tropical depression, tropical storm, hurricane, ...
- category — Saffir-Simpson scale (1–5), NA for non-hurricanes
- wind — maximum sustained wind speed, knots
- pressure — central pressure, millibars (lower = stronger)

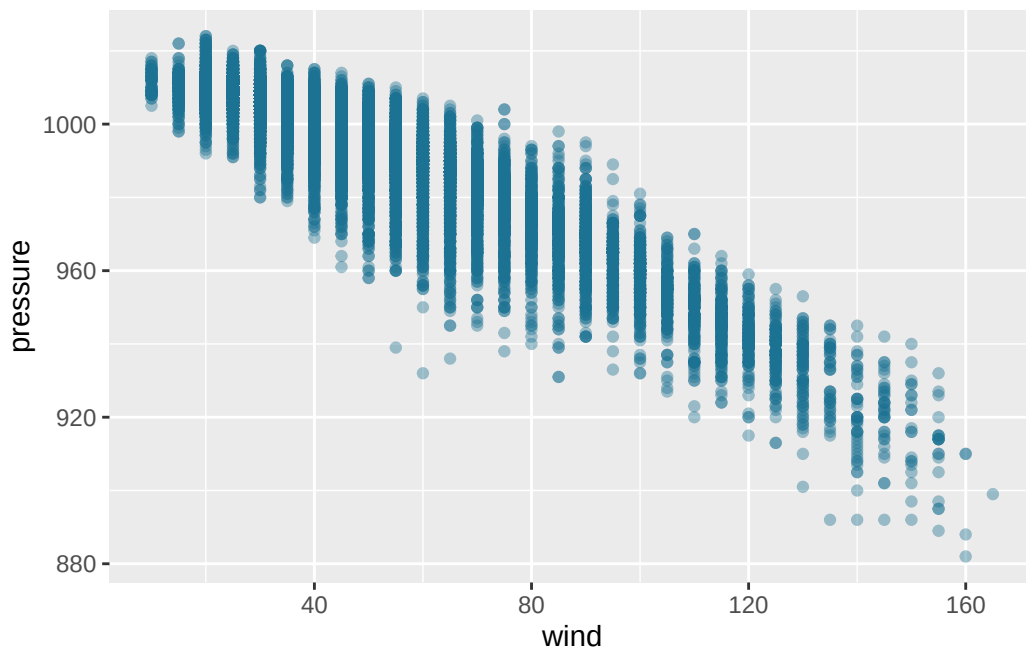
Part 1

Fill in the blanks in each chunk. Each one practices a concept from the slides.

1.1 — Mapping vs. setting

The chunk below colors every point the same shade of teal. **Without changing anything else**, modify the code so points are colored by `status` instead.

```
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(color = "#1c7293", alpha = 0.4)
```



```
# Your fixed version:
```

In **one sentence**, what's the difference between putting `color = "#1c7293"` *inside* `aes()` versus *outside* it?

[Your answer here.]

1.2 — Local mappings

The plot below colors *both* the points *and* the smoothers by `status` (so you get one smoother per status). Modify it so **only the points are colored**, but the smoother is a single curve fit to all the data.

```
ggplot(storms, aes(x = wind, y = pressure, color = status)) +  
  geom_point(alpha = 0.2) +  
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

Warning: Failed to fit group 1.

Caused by error in `smooth.construct.cr.smooth.spec()`:

! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 5.

Caused by error in `smooth.construct.cr.smooth.spec()`:

! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 6.

Caused by error in `smooth.construct.cr.smooth.spec()`:

! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 7.

Caused by error in `smooth.construct.cr.smooth.spec()`:

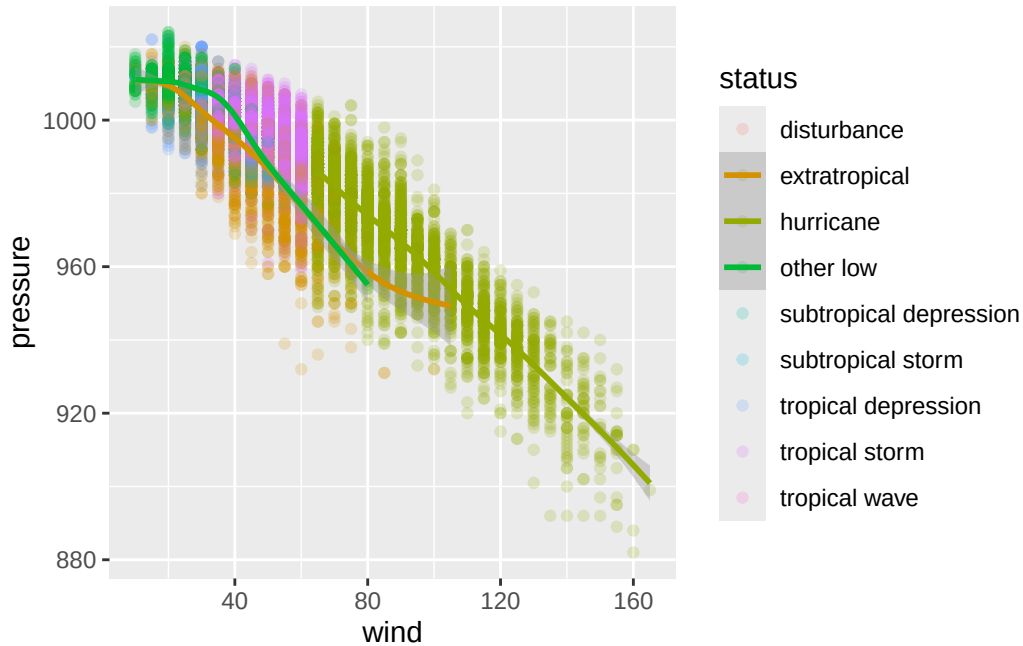
! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 8.

Caused by error in `smooth.construct.cr.smooth.spec()`:

! x has insufficient unique values to support 10 knots: reduce k.

Warning: Failed to fit group 9.
 Caused by error in `smooth.construct.cr.smooth.spec()`:
 ! x has insufficient unique values to support 10 knots: reduce k.

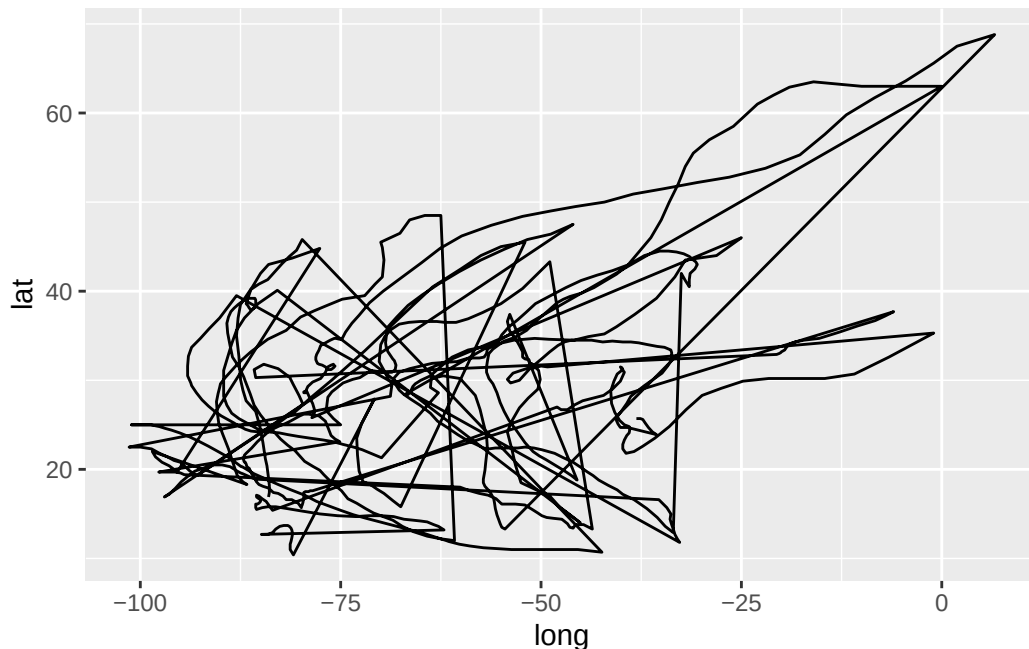


```
# Your fixed version:
```

1.3 — The group aesthetic

Run this code first. It tries to draw the path of every storm in 2005, but it produces a single zigzagging line that connects unrelated storms together.

```
ggplot(storms |> filter(year == 2005), aes(x = long, y = lat)) +  
  geom_path()
```



Add **one aesthetic mapping** so each storm gets its own path. (Hint: which variable identifies a storm?)

```
# Your fixed version:
```

1.4 — after_stat() for proportions

Make a bar chart showing the **proportion** (not count) of observations in each **status** category. You should not pre-compute the proportions with **dplyr** — let **ggplot** do it via **after_stat()**.

Hint: you'll need `group = 1`. The slide on `after_stat()` explains why.

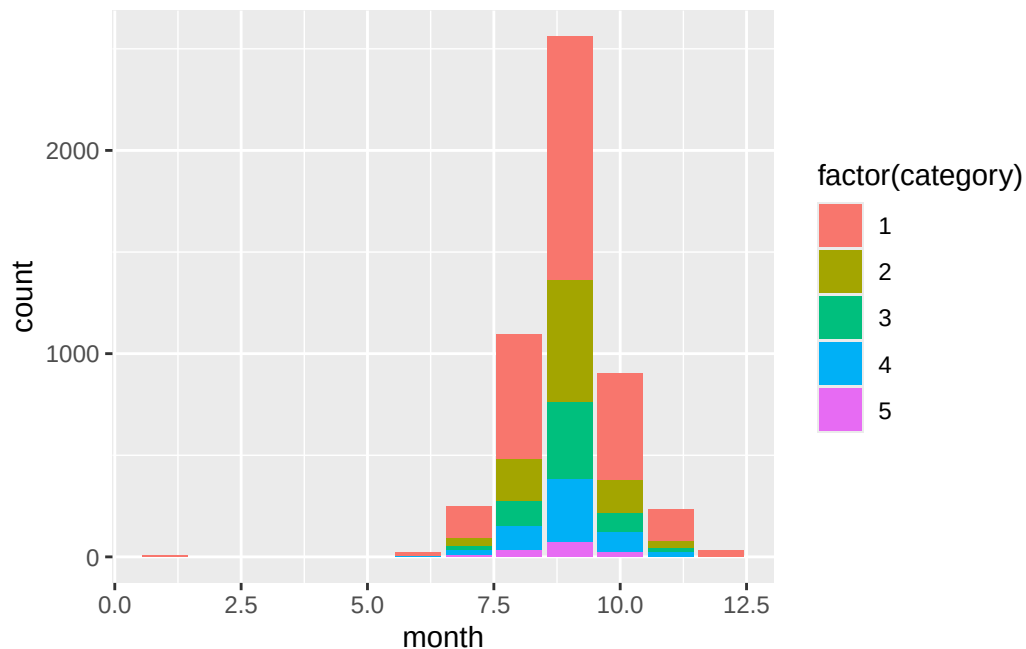
```
# Your code:
```

1.5 — Position adjustments

Run the code below. It stacks hurricane counts by category, within each month.

```
hurricanes <- storms |> filter(status == "hurricane")

ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar()
```



Make **two** more versions:

- (a) One where each bar is normalized so every month has the same height, and segments show the *proportion* of each category.
- (b) One where the categories are placed *side by side* within each month, not stacked.

```
# (a) position = "fill"
```

```
# (b) position = "dodge"
```

In **one sentence**, which of the three versions (original stacked, fill, dodge) would you use to answer the question “*In which month is Category 5 most common as a share of all hurricanes?*”

[Your answer here.]

Part 2

Pick ONE of the three questions below. Build a plot that you think answers it well, then write 2–3 sentences explaining what you see.

You're encouraged to use:

- multiple geoms in layers
- local vs. global mappings
- a stat or position adjustment we covered today
- a thoughtful color or alpha choice

You do not have to use all of them.

Question A — Wind vs. pressure

It's well-known among meteorologists that storm wind speed and central pressure have a very tight relationship. Build a plot that **shows this relationship clearly** and lets us see how **status** (or **category**) fits in.

Suggestion: layered geoms with a smoother on top, or color by category and an alpha tuned for the overplotting.

Question B — When is hurricane season?

Build a plot that shows how **hurricane activity varies through the year**. You could look at counts by month, intensity by month, or how the distribution of categories shifts as the season progresses.

Suggestion: `geom_bar()` works well here with a position adjustment, or possibly a faceted version of something simpler.

Question C — One season's storm tracks

Pick a single year (say, 2005 — a famously bad year — or any other). Build a map of **lat** vs. **long** showing the **paths of every storm** that season. Highlight one or two notable storms.

Suggestion: this needs the `group` aesthetic, and probably two `geom_path()` layers with different data.

```
# Your plot:
```

Your 2–3 sentence interpretation:

[Write here.]

Code appendix

```
library(tidyverse)
glimpse(storms)
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(color = "#1c7293", alpha = 0.4)
# Your fixed version:

ggplot(storms, aes(x = wind, y = pressure, color = status)) +
  geom_point(alpha = 0.2) +
  geom_smooth()
# Your fixed version:

ggplot(storms |> filter(year == 2005), aes(x = long, y = lat)) +
  geom_path()
# Your fixed version:

# Your code:

hurricanes <- storms |> filter(status == "hurricane")

ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar()
# (a) position = "fill"

# (b) position = "dodge"

# Your plot:
```